

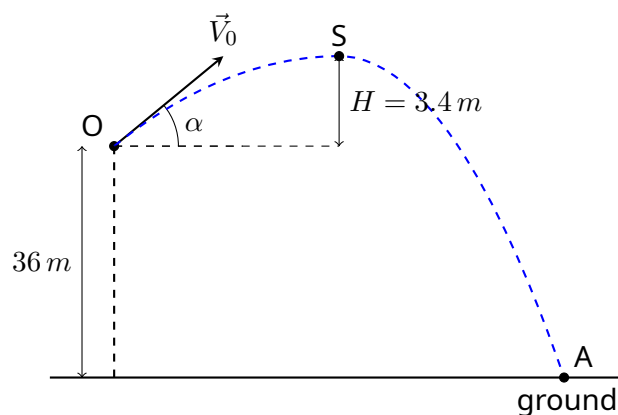
Worksheet: Work and Energy

Chapter 11 | Grade 11 Physics

Name: _____ Date: _____

Instructions: Show all formulas and substitutions clearly in your copybook. Use $g = 10 \text{ m/s}^2$ unless otherwise stated.

Exercise 1: The Volcanic Rock



In a volcanic eruption, a volcano throws a rock (R) considered as a particle, of mass $m = 20 \text{ kg}$ from a point O, 36 m above the surface of the ground. The rock is thrown with an initial velocity $\vec{V}_0$ making an angle α with the horizontal and having a magnitude of $V_0 = 24 \text{ m/s}$. S is the highest point reached by the rock, and it is at an altitude of $H = 3.4 \text{ m}$ above point O. Take the horizontal plane containing O as a reference level for gravitational potential energy and neglect air resistance.

1. The mechanical energy of the system [(R) - Earth] is conserved. Why? Calculate its value.
2.
 - 1.1) Calculate the gravitational potential energy of the system [(R) - Earth] at point S.
 - 2.2) Apply the principle of the conservation of mechanical energy to determine the speed of the rock V_S at the highest point S.
 - 3.3) Knowing that $V_S = V_0 \cos \alpha$, deduce the value of α .
3. Apply the work-energy theorem to determine the speed V_A of the rock as it reaches the ground.

Exercise 2: Vertical Launch & Energy Graphs

An object of center of mass (G) and mass $m = 2 \text{ kg}$ is launched vertically upward from the ground at $t_0 = 0$ with an initial speed V_0 . Take the horizontal plane containing (G) at $t_0 = 0$ as a reference level for gravitational potential energy. In this problem, we study the ascending motion of the object.

A) Motion without air resistance

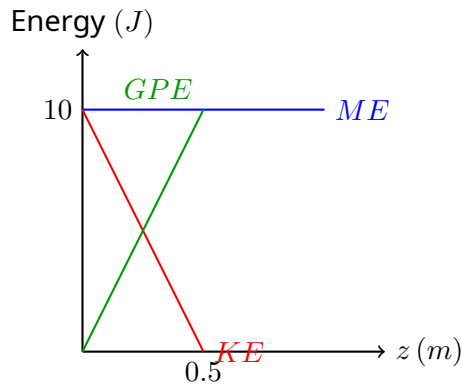


Figure (1)

The graph of figure (1) shows the mechanical energy of the system [Object – Earth] as a function of z , the altitude of (G) relative to the reference level.

1. Use figure (1) to:
 - 1.1) Show that the object is not subjected to air resistance during its ascending.
 - 1.2) Determine V_0 .
2. Write the expression of the gravitational potential energy of the system [Object – Earth] as a function of z , and then describe its curve on figure (1).
3. Determine the expression for the kinetic energy of the object as a function of z .
4. Refer to the expressions (or the graph concepts) to determine the maximum altitude Z_{max} reached by (G).

B) Motion with air resistance

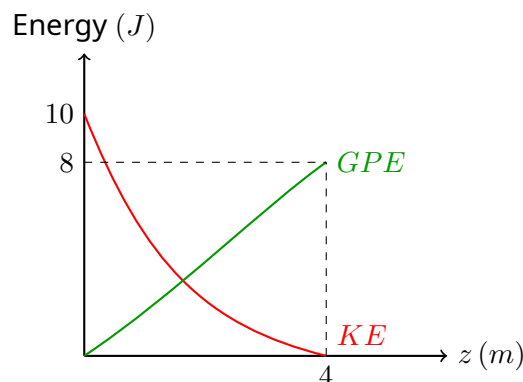
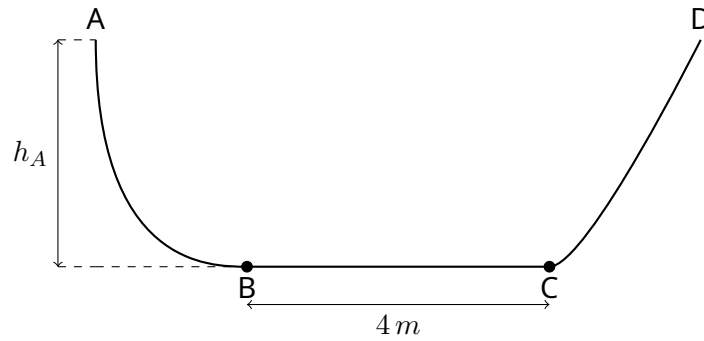


Figure (2)

Suppose that, during its ascending, the object is subjected to air resistance which is constant and directed vertically downward. The graph of figure (2) represents the kinetic energy of the object and the gravitational potential energy of the system [Object – Earth] as a function of z .

1. Specify which curve represents the variation of the kinetic energy. Justify.
2. Determine the maximum height Z_{max} reached by (G).
3. Determine the amount of thermal energy produced during the ascending of the object.

Exercise 3: The Skier



Consider a skier taken as a particle moving along a path ABCD in the vertical plane. The mass of the skier and his equipment is $M = 60 \text{ kg}$. Along the track BC, whose length is 4 m , the skier is acted upon by a constant friction force of magnitude $f = 60 \text{ N}$. The track CD is frictionless. The horizontal plane containing BC is taken as a reference level for gravitational potential energy. The skier starts from rest from point A at an altitude $h_A = 3.5 \text{ m}$ above the track BC.

1. Along the track AB

- 1.1) Determine the mechanical energy of the system Skier-Earth when the skier is at point A.
- 1.2) The skier reaches point B with a speed $V_B = \sqrt{70} \text{ m/s}$. Prove that the track AB is frictionless.

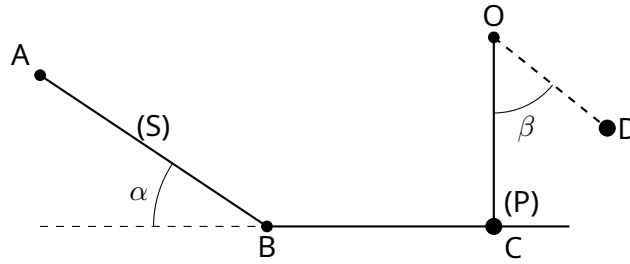
2. Along the track BC

- 2.1) Determine the work done by each of the external forces acting on the skier between B and C.
- 2.2) Deduce the speed of the skier at C.
- 2.3) The kinetic energy lost between B and C is converted into another form of energy. Name it.

3. Along the track CD

- 3.1) Calculate:
 - 3.1.1) The mechanical energy of the system Skier-Earth when the skier is at point C.
 - 3.1.2) The gravitational potential energy of the system Skier-Earth when the skier is at point D (assume $h_D = 3.5 \text{ m}$).
 - 3.2) Deduce that the skier cannot reach point D.
 - 3.3) In reality, I is the highest point the skier can reach before returning back. Determine the height h_I of point I above the track BC.
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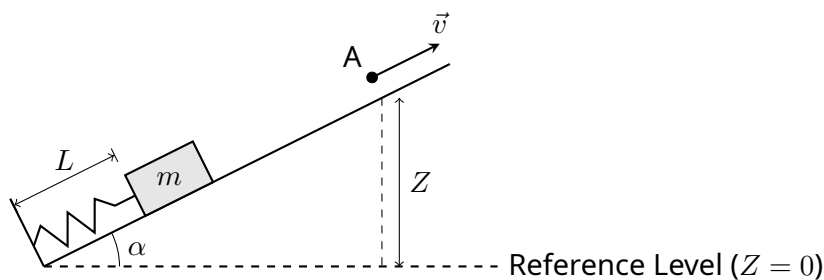
Exercise 4: The Track & Collision



An object (S), taken as a particle of mass $m = 2\text{ kg}$, is released from rest at a point A of the track ABC. Given: $\sin(\alpha) = 0.2$; $\cos(\alpha) = 0.98$; $AB = 4\text{ m}$ and BC is frictionless. The horizontal plane containing BC is taken as a reference level for gravitational potential energy.

1. 1.1) (S) reaches a point B with a speed $V_B = 3\text{ m/s}$. Determine the magnitude of the constant friction force exerted by the track AB on (S).
- 1.2) Determine the instantaneous power of: the friction force, normal force and the weight of (S), when the speed of (S) becomes 2 m/s .
- 1.3) It takes the object 2.7 s to move from A to B. Deduce the average power of the friction force acting on (S) between A and B.
2. The object (S) continues its motion and reaches a point C, where it collides with another object (P) of mass $M = 3\text{ kg}$, attached to one end of an inextensible light vertical cord of length $\ell = 50\text{ cm}$. The other end of the cord is attached to a fixed point O. Right after collision between (S) and (P), object (P) is ejected with a speed of $V_0 = 1.6\text{ m/s}$. When (P) reaches point D, the cord is deviated by a maximum angle β . Determine the value of β .
3. As object (P) reaches point D, the cord is cut and (P) falls vertically downwards.
 - 3.1) Determine the mechanical energy of the system [(P) – Earth] at point D.
 - 3.2) Determine the speed of (P) and its altitude above ground when its kinetic energy becomes equal to the gravitational potential energy of the system [(P) – Earth].
 - 3.3) Determine the speed V_{ground} of (P) upon reaching the ground.

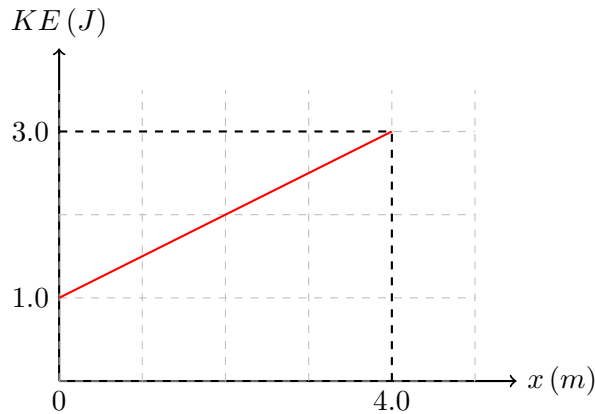
Exercise 5: Spring Launch



To launch a solid of mass m on an inclined slide, we use the apparatus shown in the figure above. The natural length of the spring is L_0 . Before launching, the spring is compressed to a length L . After launching, the center of mass of the solid reaches the point A of altitude Z with a velocity v . Neglect friction.

- Determine the relation among L_0, L, m, v, Z, g , and the constant k of the spring.
 - Calculate k if the maximum altitude reached is 20 cm . Take $m = 600\text{ g}$; $g = 10\text{ m/s}^2$; $L_0 = 25\text{ cm}$; $L = 20\text{ cm}$.
 - Determine the instantaneous power of the weight of the solid at point A.
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Exercise 6: Inclined Plane & Energy Graph



A solid S of mass $m = 0.2\text{ kg}$ slides down a rough inclined plane making an angle $\alpha = 30^\circ$ with the horizontal along the line of greatest slope ox . The frictional force is assumed to be constant with magnitude f . A device is used to plot the variations of the kinetic energy as a function of the abscissa x of the solid. Knowing that the solid is at the origin O ($x = 0$) at $t = 0$, refer to the graph above to answer the following:

- Calculate the initial speed V_0 of the solid at $t = 0$.
- What is its speed after it has traveled a distance $x = 4\text{ m}$?
- What is the sum $\Sigma W(\vec{F}_{ext})$ of the works of all external forces applied to the solid after a displacement $OA = x_A = 4\text{ m}$ on the inclined plane?
- Write the algebraic expression of $\Sigma W(\vec{F}_{ext})$ in terms of x_A, g, m, f , and $\sin \alpha$, then deduce the value of the friction force f .
- Calculate the instantaneous power dissipated by the friction force at $x = 4\text{ m}$.
- Deduce the average power of the friction force during the displacement OA, knowing this motion lasts $\Delta t \approx 0.92\text{ s}$.